

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2024-2025

Comprehensive Examination
(EC-3 Regular)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 29/03/2025 (EN)

No. of Pages	= 3
No. of Questions	= 10

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. A healthcare data analytics application utilizes a Cloud-based PostgreSQL cluster. The analytics team conducts patient record analysis on the data stored on the Cloud instance. It takes approximately 30 seconds per user query on average. You are tasked to implement a Memcached based cache in the healthcare center's data center to expedite these queries, aiming for an average query latency of 10 seconds. The Memcached cache has an access latency of 5 seconds. When the cache is established, data can only be transferred from the cache to the application and not directly from the Cloud DB. What should be the projected hit rate in the cache to achieve this ambitious average query latency target?
What type of locality of reference is used here? [Marks 4]

Q.2. A company decided to perform real time analytics on its Web server logs to gain insights on website visitors, behavior, crawlers accessing the site, business insights, security issues, and more. For this, the log file entries are to be streamed to an HDFS folder in a Hadoop cluster. Streaming the logs to HDFS is to be done only when 1000 or more entries are made to the webserver log. Design a Flume agent to implement the streaming with the webserver log as the source and HDFS folder as the sink. [Marks 4]

Q.3. Describe the most appropriate scheduling mechanism to be configured with Hadoop YARN in the following scenarios:
(a). A company wants to run jobs of equal size on a Hadoop cluster. The number of jobs can vary at any point in time.
(b). A company wants to run short jobs and long jobs on the same cluster and short jobs cannot always be starved by long jobs.
(c) A company wants to divide the Hadoop cluster resources in different proportions between the departments and does not care about wastage of unutilized resources.
(d) A company wants to run jobs of different nature concurrently on the same cluster and is worried about wastage of resources [Marks 4]

- Q.4. A retail company stores its transactional data in a MySQL database. The data science team wants to perform analytics using Hive on Hadoop. They need load the historic data and continue to load the new, updated records.
- How would you use Sqoop to achieve this?
 - Which Sqoop import mode is best suited for this task?
 - What considerations should be taken to ensure efficient incremental loading?
- (Also consider other Hadoop tools) [Marks 4]

- Q.5. Data stored on an SQL database is normalized into 3 tables (Users, Professions, Cars) as shown below:

Users

User_id	First_Name	Surname	Mobile	City	Location_x	Location_y
1	Paul	Miller	4085575051	London	45.123	47.232

Professions

ID	User_id	Profession
10	1	Banking
11	2	Finance
12	3	Trader

Cars

ID	User_id	Model	Year
20	1	Bentley	1993
21	2	Rolls Royce	1998
22	3	BMW	2005

Model the above data in MongoDB for the given user. When you are designing your MongoDB document structure, make sure that it will give optimum query performance for your application. Select a suitable value for the field `_id` of the document.

[Marks 4]

- Q.6. A spark cluster is to be configured to process a file of size 48 GB. The processing is of CPU intensive nature. You are asked to configure the cluster on a 5-node cluster in which 1 node will have to be used as the driver and the other 4 nodes will have to be used as worker nodes. Each of the nodes of the cluster is having 16 Gib memory and 16 core CPUs. On each of the nodes, you need to reserve 4Gib for the operating system and other applications. The rest of the memory can be used for Spark. On each of the worker nodes, you need reserve 4 cores for the operating system and other processes. For optimum performance, provide values for the following:
1. Number of RDD partitions
 2. Number of executors
 3. Number of cores and size of memory allocated to each executor
 4. Number of tasks getting executed on each executor

[Marks 4]

Q.7. Go through the Spark Scala code given below and answer the questions (a,b,c,d) given below:

```
1  val lines = new Array[String](2)
2  lines(0) = "one one reject"
3  lines(1) = "own own own"
4  val stringRDD=sc.parallelize(lines,2)
5  println("Number of Partitions: "+stringRDD.getNumPartitions)
6  val wordsRDD = stringRDD.flatMap(x => x.split(" "))
7  val wc = wordsRDD.map(word => (word, 1)).reduceByKey(_ + _)
8  wc.collect()
9  val filt1RDD = wordsRDD.filter(x => x.contains("own"))
10. filt1RDD.collect()
```

Questions:

- What is the output at line No. 5
- What is the output at line No. 8
- What is the output at line No.10
- Modify the code to find out the word count after excluding the word "reject"

[Marks 4]

Q.8. Differentiate between wide and narrow transformations in Apache Spark. Which transformations trigger shuffling, and how does it impact RDDs? Discuss best practices to optimize shuffling in Spark.

[Marks 4]

Q.9. Why is windowing needed in Stream computing? Data from an IOT device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 event per second. The batching interval in the streaming context is set to 10 seconds. The stream analytics system uses a stateful algorithm with an analysis window of 120 seconds. It is required to generate an output from the streaming system every 60 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the Window suggested by you. What will be the number of RDDs in the Dstreams generated in the window duration.

[Marks 4]

Q.10. You are working for a company that is building an e-commerce platform, and they are considering using DynamoDB in Provisioned mode. They anticipate the following workload requirements:

- Read Operations: 50,000 strongly consistent reads per second, each reading 8KB items.
- Write Operations: 20,000 transactional writes per second, each writing 3KB of data.

Calculate the number of RCU (Read Capacity Units) and WCU (Write Capacity Units)?

[Marks 4]
